Isbarov Jafar

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Citizenship: Azerbaijan

Research interests

Natural language processing, trustworthy machine learning, single-cell analysis, systems biology

Education

The George Washington University — Washington, D.C., the US
M.Sc. Computer Science — Sep, 2023 – Present

Azerbaijan State Oil and Industry University — Baku, Azerbaijan
B.Sc. Biomedical Engineering — Sep, 2018 – Jul, 2023

Tianjin University — Tianjiin, China
Foundation degree — Sep, 2017 – Jul, 2018

Research output

Comparison of algorithms used in single-cell transcriptomic data analysis
Jafar Isbarov, Elmir Mahammadov
PREPRINT
Elmi Spektr Tutorship Programme, 2021.
DOI: http://dx.doi.org/10.13140/RG.2.2.32644.71045

Robust Automated Spelling Correction with Deep Ensembles
Jafar Isbarov, Samir Rustamov, Kavsar Huseynova
Presented in ACM ISMSI 2024
http://ceferisbarov.github.io/files/ISMSI_final.pdf

Experience

PRODATA LLC — Baku, Azerbaijan
Machine learning engineer — March, 2023 –
Pre-trained foundation models for Azerbaijani language, fine-tuned them for downstream tasks. Developed an on-premise RAG (retrieval-augmented generation) system for Azerbaijani. Deployed LLMs via FastAPI and Docker. Automated the development process via git, GitHub Actions and Docker Hub. Involved in development of a GUI for chatbot development. Developed a custom container orchestrator for the said project.

Azerbaijan Space Agency — Baku, Azerbaijan
Machine learning engineer — July, 2022 – March, 2023

Data scraping and processing for NLP problems. Built machine Learning models in sklearn and TensorFlow. Deploying the models via FastAPI and Docker. Developed a spelling correction solution for Azerbaijani language.

Elmi Spektr Nijmegen, Netherlands
Research intern Jan, 2021 – May, 2021
Comparative analysis of clustering and dimensionality reduction algorithms used single-cell transcriptomic analysis.

Skills

Programming
Proficient in: Python, Julia.
Familiar with: C++, SQL.
Frameworks: TensorFlow, PyTorch, Scikit-learn, FastAPI
Databases: MongoDB, MySQL, Oracle
Other: git, Docker, Linux

Industry knowledge
Natural language processing, MLOps, Backend development

Languages
English (C2), Azerbaijani (C2), Chinese (B2), Turkish (B2)

Certifications

IELTS Academic
Reading: 9.0, Listening: 9.0, Speaking: 8.0, Writing: 7.0, Overall: 8.5

GRE
Quantitative: 164, Verbal: 165, Writing: 4.0

TensorFlow Developer Certificate

Service and outreach

Elmi Spektr May, 2021 – Present
Elmi Spektr is an independent platform that engages with issues related to public scientific education. I am one of the main content creators of the platform. I also provide technical support when necessary. One of the tools that I developed for internal use later became an open source project.

Ecofront April, 2022 – Present
Eco-activist group fighting against the crisis of environment, biodiversity and climate. I participate in various awareness-raising activities, and sometimes overtake managerial duties.

Other interests

Ethical philosophy, classical literature, movies, music, gaming, fitness, football.